

Research Reflection - Senior Design – PolyExposure Control Rebuild Project (Taylor Corporation)

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My senior design project, conducted in partnership with Taylor Corporation, represents the most professionally immersive research experience of my academic career. The PolyExposure Control Rebuild Project focuses on redesigning and modernizing the control system used in a poly exposure manufacturing process. Unlike traditional academic research, this project directly impacts real industrial operations, meaning that our design decisions carry real operational, safety, and financial consequences.

When I first entered the project alongside my teammates Nathnael Minuta and Mustanser Ahmad, I initially approached it with the mindset of a student completing a course requirement. That mindset changed quickly. From our first interactions with Taylor engineers, it became clear that this was not simply a classroom assignment. We were working with legacy industrial systems that had been operating for years, meaning that our responsibility was not just to create something new, but to understand why the existing system was designed the way it was before attempting to improve it.

Our project was guided by two mentors who played complementary roles in shaping our experience. Professor Bhushan, the faculty advisor responsible for the senior design course, guided us from the academic side by ensuring we approached the project with the discipline expected of professional engineers. He regularly checked in on our progress, sometimes even attending our project meetings to help us analyze challenges. Early in the semester, he told us something that stayed with me throughout the project: that we were not only completing our own senior design project, but also representing the university and setting the stage for how future student teams would be perceived by Taylor Corporation. He emphasized that strong engineers do not simply fix problems quickly; they take the time to investigate the root causes of issues deeply before proposing solutions. This mindset encouraged us to slow down our initial assumptions and focus on understanding the system thoroughly before proposing changes.

On the industry side, we worked closely with Colin, a senior staff engineer at Taylor Corporation who served as our primary technical mentor. Colin played a critical role in helping us understand the practical realities of the system we were working with. In the early stages of the project, our team began by studying the electrical wiring and hardware layout of the existing exposure system. At that time, we did not yet have a complete understanding of how the system functioned as a whole. Colin spent hours walking us through the system architecture, patiently explaining how different subsystems interacted and why certain design decisions had been made in the past.

One of the earliest pieces of advice he gave us was that effective engineers must always understand the full picture of what they are building. Without that system-level understanding, even technically correct decisions can lead to unintended consequences. This advice significantly influenced how we approached the rest of the project. We would always try to ask a lot of questions and get a full understanding in order to deliver a quality work.

After developing our initial understanding of the system, we began designing the first iteration of updated electrical schematics. When we presented our first design to Colin, the feedback we received was both detailed and constructive. Several critical components had been overlooked, including safety relay integration, signal isolation considerations, and certain interlock conditions required for safe machine operation. While it was initially discouraging to realize how much we had missed, Colin reassured us that this type of iteration is a normal part of professional engineering design.

We revised the schematics based on his feedback and presented a second iteration. This version addressed many of the earlier concerns, but additional improvements were suggested, such as clearer labeling of signal paths and better documentation of system dependencies. Over the course of the project, we repeated this design–review–revision cycle four times. Each iteration improved the clarity, safety, and practicality of our design. Experiencing this process firsthand helped me understand that engineering design rarely happens in a single attempt; it evolves through continuous review and refinement.

Once our electrical schematics were approved, we transitioned to another major component of the project: PLC design and control logic implementation. This presented a new challenge because none of us had significant prior experience with PLC programming in an industrial automation context. Instead of seeing this as a limitation, we treated it as an opportunity to learn.

With guidance from both Professor Bhushan and Colin, we began researching PLC architecture and industrial automation programming practices. We installed the PLC emulator environment, studied technical documentation, and reviewed instructional materials they provided. We also watched tutorials and analyzed example control logic implementations to better understand how industrial control systems are structured.

As we began developing the control logic, the same iterative process we experienced with the electrical schematics repeated itself. Our early PLC logic designs often worked conceptually but failed to account for edge cases such as unexpected operator inputs, timing dependencies, or safety interlocks. When we presented these designs during review meetings with Colin, he often asked questions that forced us to reconsider our assumptions. For example, he would ask what would happen if a sensor failed, if an operator attempted to trigger a process out of sequence, or if two subsystems attempted to activate simultaneously. These discussions pushed us to think more deeply about robustness and safety in control systems.

Through this process, I learned that professional engineering design involves anticipating failure scenarios as much as designing for normal operation. Each revision improved not only our technical implementation but also our understanding of how industrial automation systems must behave reliably under a wide range of conditions.

One of the most valuable learning experiences occurred when we presented our progress to Taylor engineers and received direct feedback on our design approach. During one of these presentations, we explained our proposed modernization strategy and the reasoning behind our control logic structure. The engineers asked several questions that challenged our assumptions, particularly regarding maintainability and long-term serviceability of the system. One engineer pointed out that while our design was technically sound, it would also need to be understandable and maintainable by technicians who might work on the machine years later. This feedback reinforced an important lesson: engineering solutions must consider not only technical performance but also usability, documentation quality, and long-term operational sustainability.

The project also required strong collaboration within our team. Nathnael, Mustanser, and I each approached engineering problems differently. Nathnael often focused on structural and system integration clarity. Mustanser approached problems with analytical precision and modeling perspective. I often focused on system functionality, control logic behavior, and implementation feasibility. Learning to integrate these perspectives strengthened our design outcomes and required us to constantly communicate and align technical decisions.

Another significant challenge involved learning to work within real industrial constraints. Unlike academic projects that prioritize optimization, industry systems must balance cost, maintainability, downtime risk, operator usability, and regulatory safety requirements. Many design ideas that initially seemed technically superior became impractical when evaluated within these constraints. Learning to evaluate engineering solutions within operational reality became one of the most transformative aspects of the project.

This senior design experience fundamentally reshaped my understanding of research as professional engineering practice.

One of the most significant competencies I developed was investigative research. Unlike the power cell project, where we built new solutions from scratch, this project required understanding existing systems deeply before attempting improvement. I learned that modernization research often involves historical analysis, system interpretation, and diagnostic reasoning. This form of research requires patience, curiosity, and respect for prior engineering decisions.

The project also strengthened my ability to conduct research within complex constraint environments. Academic research often emphasizes innovation, but industrial research demands balance between innovation and practicality. I learned how to evaluate solutions not only for performance improvement, but also for safety, cost efficiency, maintainability, and user interaction. This expanded my research perspective from technical optimization to systems sustainability.

Another major area of growth involved professional communication and stakeholder interaction. Presenting research findings to Taylor engineers required clarity, technical justification, and adaptability to feedback. I learned that professional research is not simply about being correct. It is about being explainable, defensible, and collaborative. This experience strengthened my confidence in presenting technical ideas while remaining open to revision.

The project also deepened my understanding of accountability in engineering research. Knowing that our recommendations could influence real production systems introduced a level of pressure I had never experienced before. This pressure forced me to adopt higher standards for documentation, verification, and validation. I became more meticulous in design review, more cautious in assumptions, and more responsible in technical decision making.

Working in a multidisciplinary environment further strengthened my collaborative research skills. Control system modernization required integration of electrical engineering, embedded programming, automation design, and human-machine interface development. This reinforced my understanding that modern engineering research is inherently interdisciplinary and that effective researchers must communicate across technical boundaries.

Perhaps the most significant transformation was identity-based. Throughout the project, I began noticing a shift in how I approached engineering challenges. I stopped asking, "What is the correct answer?" and started asking, "What is the most responsible solution?" That shift represents a transition from student learning to professional engineering thinking.

Moving forward, I intend to pursue engineering roles and research opportunities that involve system modernization, embedded systems integration, and industrial automation. This project confirmed that I am most motivated when working on engineering challenges that directly influence real-world operations. I also plan to continue developing my ability to communicate technical research effectively within professional environments and to approach engineering work with the accountability, safety awareness, and systems thinking this project helped develop.